

Problem 3

Problem 1 Circle the correct answer in each part:

(a) Consider the limit $\lim_{x \rightarrow 0} (\cos(x))^{\sin(x)}$.

(i) Evaluate the limit.

i. The limit DNE

ii. e

iii. 1

iv. ∞

v. $-\infty$

vi. 0

vii. None of the previous answers is correct

(ii) What Limit Law, rule or technique did you use to find this limit?

i. The Squeeze Theorem;

ii. L'Hôpital's Rule;

iii. The Product Law;

iv. Evaluated the function at $x = 0$, since the function is continuous at $x = 0$;

v. None of the previous answers is correct

(b) Evaluate the limit $\lim_{x \rightarrow 4^-} \frac{\ln(x)}{x - 4}$.

(i) The limit DNE

(ii) e

(iii) 1

(iv) ∞

(v) $-\infty$

(vi) 0

(vii) None of the previous answers is correct

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(c) Evaluate the limit $\lim_{x \rightarrow \infty} \frac{\ln(x)}{x-4}$.

- (i) The limit DNE
- (ii) e
- (iii) 1
- (iv) ∞
- (v) $-\infty$
- (vi) 0
- (vii) None of the previous answers is correct

(d) Consider the limit

$$\lim_{h \rightarrow 0} \frac{(2+h)^3 - 8}{h} = f'(2).$$

Determine the function f .

- (i) Such a function DNE;
- (ii) $f(x) = x^3$;
- (iii) $f(x) = (2+x)^3$;
- (iv) $f(x) = \frac{(2+x)^3}{x}$;
- (v) None of the previous answers is correct

(e) Consider the limit:

$$\lim_{x \rightarrow 0^+} \left(\frac{\sin(x)}{x} \right)^{|\ln(x)|} dt.$$

Determine the form of this limit.

- (i) $\frac{0}{0}$;
- (ii) $\frac{\infty}{\infty}$;
- (iii) 1^0 ;
- (iv) 0^0 ;
- (v) 1^∞ ;
- (vi) ∞^∞ ;
- (vii) None of the previous answers is correct